

Norway

SLEIPNER

Volve F

F-14

F-14

Design: F-14

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Inspirer Rotary Table @ 54.90m (Actual Rotary Table)
Site:	Volve F	MD Reference:	Inspirer Rotary Table @ 54.90m (Actual Rotary Table)
Well:	F-14	North Reference:	Grid
Wellbore:	F-14	Survey Calculation Method:	Minimum Curvature
Design:	F-14	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude: 58° 26' 29.8068 N
From:	Map	Easting:	435,050.02 m	Longitude: 1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence: -0.95 °

Well	F-14			
Well Position	+N/-S	-1.21 m	Northing:	6,478,562.31 m
	+E/-W	2.42 m	Easting:	435,052.44 m
Position Uncertainty	0.00 m	Wellhead Depth:	91.00 m	Water Depth: 91.00 m

Wellbore	F-14				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MNETICREFERENCE	04.11.2007	-2.23	71.63	50,339

Design	F-14			
Audit Notes:				
Version:	1.0	Phase:	ACTUAL	Tie On Depth: 145.90
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)
	145.90	-1.21	2.42	61.23

Survey Program	Date	30.09.2017		
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description
155.09	2,655.90	F-14 12 1/4" Drop Gyro (F-14)	Wellbore Surveyor, stat	Gyro Tool from GD
2,655.90	2,788.00	F-14 12 1/4" MWD Survey (F-14)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)
2,788.00	3,729.86	F-14 8 1/2" MWD Survey (F-14)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	-1.21	2.42	0.00	0.000	0.000	0.000	
155.09	0.18	5.42	155.09	-1.20	2.42	0.01	0.588	0.588	0.000	
195.44	0.33	37.78	195.44	-1.04	2.50	0.15	0.150	0.112	24.059	
235.80	0.61	58.22	235.80	-0.84	2.75	0.47	0.239	0.208	15.193	
276.20	1.55	62.50	276.19	-0.47	3.42	1.23	0.700	0.698	3.178	
316.60	1.55	67.03	316.58	-0.01	4.41	2.32	0.091	0.000	3.364	
356.90	1.90	71.62	356.86	0.42	5.54	3.52	0.280	0.261	3.417	

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Wellbore:	F-14	Survey Calculation Method:	Minimum Curvature
Design:	F-14	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
397.30	3.61	88.22	397.21	0.67	7.45	5.31	1.388	1.270	12.327	
437.70	4.31	83.62	437.51	0.87	10.23	7.85	0.570	0.520	-3.416	
478.00	4.00	69.33	477.71	1.54	13.05	10.64	0.802	-0.231	-10.638	
518.40	3.37	50.31	518.03	2.79	15.28	13.20	1.015	-0.468	-14.124	
558.70	2.15	26.90	558.28	4.23	16.53	14.99	1.219	-0.908	-17.427	
599.10	0.79	14.51	598.67	5.17	16.95	15.81	1.031	-1.010	-9.200	
639.50	0.26	72.65	639.06	5.47	17.10	16.09	0.512	-0.394	43.173	
679.80	0.26	75.13	679.36	5.52	17.28	16.27	0.008	0.000	1.846	
720.20	0.27	77.66	719.76	5.56	17.46	16.45	0.011	0.007	1.879	
760.50	0.29	85.64	760.06	5.59	17.66	16.63	0.033	0.015	5.940	
800.90	0.35	107.43	800.46	5.56	17.88	16.81	0.100	0.045	16.181	
841.30	0.47	160.22	840.86	5.37	18.05	16.87	0.282	0.089	39.200	
881.70	0.48	160.60	881.26	5.05	18.16	16.82	0.008	0.007	0.282	
922.00	0.35	168.30	921.56	4.77	18.24	16.75	0.105	-0.097	5.732	
962.40	0.26	163.24	961.96	4.56	18.29	16.70	0.070	-0.067	-3.757	
1,002.80	0.22	163.51	1,002.36	4.40	18.34	16.66	0.030	-0.030	0.200	
1,043.20	0.23	170.57	1,042.76	4.25	18.38	16.62	0.022	0.007	5.243	
1,083.50	0.26	215.79	1,083.06	4.09	18.34	16.51	0.142	0.022	33.663	
1,123.90	0.26	211.34	1,123.46	3.94	18.24	16.35	0.015	0.000	-3.304	
1,164.20	0.20	193.67	1,163.76	3.79	18.17	16.22	0.069	-0.045	-13.154	
1,204.60	0.19	177.78	1,204.16	3.66	18.16	16.14	0.041	-0.007	-11.800	
1,245.00	0.19	174.09	1,244.56	3.53	18.17	16.09	0.009	0.000	-2.740	
1,285.40	0.19	173.14	1,284.96	3.39	18.18	16.04	0.002	0.000	-0.705	
1,325.80	0.19	171.51	1,325.36	3.26	18.20	15.99	0.004	0.000	-1.210	
1,366.10	0.18	125.54	1,365.66	3.16	18.26	15.99	0.108	-0.007	-34.221	
1,406.50	0.18	123.89	1,406.06	3.08	18.37	16.05	0.004	0.000	-1.225	
1,446.80	0.18	125.43	1,446.36	3.01	18.47	16.10	0.004	0.000	1.146	
1,487.20	0.18	125.43	1,486.75	2.94	18.57	16.16	0.000	0.000	0.000	
1,527.50	0.18	127.74	1,527.05	2.86	18.68	16.21	0.005	0.000	1.720	
1,567.90	0.18	129.47	1,567.45	2.78	18.77	16.26	0.004	0.000	1.285	
1,608.20	0.18	129.69	1,607.75	2.70	18.87	16.31	0.001	0.000	0.164	
1,648.60	0.19	223.96	1,648.15	2.62	18.87	16.27	0.201	0.007	70.002	
1,689.00	0.19	223.56	1,688.55	2.52	18.78	16.14	0.001	0.000	-0.297	
1,729.30	0.19	225.71	1,728.85	2.42	18.69	16.01	0.005	0.000	1.600	
1,769.70	0.19	224.86	1,769.25	2.33	18.59	15.88	0.002	0.000	-0.631	
1,810.00	0.19	226.63	1,809.55	2.24	18.50	15.75	0.004	0.000	1.318	
1,850.40	0.19	227.97	1,849.95	2.14	18.40	15.62	0.003	0.000	0.995	
1,890.80	0.19	260.83	1,890.35	2.09	18.28	15.50	0.080	0.000	24.401	
1,931.10	0.19	261.93	1,930.65	2.07	18.15	15.37	0.003	0.000	0.819	
1,971.50	0.19	263.57	1,971.05	2.05	18.02	15.25	0.004	0.000	1.218	
2,011.80	0.19	258.37	2,011.35	2.03	17.89	15.12	0.013	0.000	-3.871	
2,052.50	0.19	238.20	2,052.05	1.98	17.76	14.99	0.049	0.000	-14.867	

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Wellbore:	F-14	Survey Calculation Method:	Minimum Curvature
Design:	F-14	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
2,092.50	0.16	275.09	2,092.05	1.95	17.65	14.88	0.086	-0.022	27.667	
2,132.90	1.74	12.52	2,132.45	2.56	17.73	15.23	1.313	1.173	72.349	
2,173.20	3.98	16.68	2,172.69	4.49	18.26	16.63	1.674	1.667	3.097	
2,213.60	7.98	18.75	2,212.86	8.49	19.57	19.70	2.974	2.970	1.537	
2,253.90	8.15	20.38	2,252.77	13.82	21.46	23.93	0.212	0.127	1.213	
2,294.30	8.34	21.84	2,292.75	19.22	23.55	28.36	0.210	0.141	1.084	
2,334.60	10.09	24.04	2,332.53	25.16	26.07	33.43	1.329	1.303	1.638	
2,375.00	12.41	27.18	2,372.15	32.26	29.50	39.85	1.781	1.723	2.332	
2,415.30	14.84	28.10	2,411.31	40.66	33.91	47.76	1.816	1.809	0.685	
2,455.80	17.20	31.13	2,450.24	50.36	39.45	57.28	1.854	1.748	2.244	
2,496.10	19.35	41.69	2,488.51	60.45	46.97	68.73	2.933	1.600	7.861	
2,536.50	23.77	49.56	2,526.08	70.74	57.63	83.02	3.916	3.282	5.844	
2,576.90	28.22	56.01	2,562.40	81.37	71.75	100.52	3.911	3.304	4.790	
2,617.30	32.30	55.37	2,597.28	92.85	88.56	120.78	3.039	3.030	-0.475	
2,655.90	35.58	54.59	2,629.30	105.22	106.20	142.20	2.572	2.549	-0.606	
2,673.90	37.39	54.82	2,643.77	111.40	114.94	152.83	3.025	3.017	0.383	
2,706.97	39.85	55.29	2,669.61	123.22	131.86	173.35	2.247	2.232	0.426	
2,747.16	40.19	55.74	2,700.39	137.86	153.16	199.07	0.333	0.254	0.336	
2,790.39	39.56	54.98	2,733.57	153.61	175.96	226.64	0.553	-0.437	-0.527	
2,830.75	43.81	57.30	2,763.70	168.54	198.26	253.37	3.360	3.159	1.724	
2,870.67	49.28	59.12	2,791.15	183.78	222.89	282.29	4.229	4.111	1.368	
2,911.52	54.40	59.01	2,816.38	200.29	250.43	314.38	3.761	3.760	-0.081	
2,950.80	60.08	58.76	2,837.63	217.35	278.70	347.37	4.341	4.338	-0.191	
2,991.49	64.34	59.52	2,856.59	235.81	309.59	383.33	3.180	3.141	0.560	
3,032.17	66.85	65.62	2,873.41	252.84	342.46	420.34	4.495	1.851	4.499	
3,072.79	69.70	71.10	2,888.46	266.73	377.52	457.76	4.308	2.105	4.047	
3,112.44	71.90	73.97	2,901.50	277.96	413.23	494.47	2.641	1.665	2.172	
3,153.53	74.71	74.92	2,913.30	288.51	451.15	532.78	2.156	2.052	0.694	
3,193.66	77.75	78.46	2,922.86	297.48	489.07	570.33	3.431	2.273	2.646	
3,233.21	80.75	81.97	2,930.23	304.07	527.35	607.06	3.467	2.276	2.662	
3,273.30	81.08	85.25	2,936.57	308.48	566.68	643.66	2.436	0.247	2.454	
3,314.97	76.91	85.46	2,944.52	311.79	607.44	680.99	3.006	-3.002	0.151	
3,355.37	71.94	85.70	2,955.36	314.79	646.23	716.43	3.695	-3.691	0.178	
3,395.93	71.34	85.44	2,968.14	317.76	684.61	751.50	0.480	-0.444	-0.192	
3,436.15	67.98	86.53	2,982.12	320.41	722.22	785.75	2.620	-2.506	0.813	
3,476.50	65.26	89.03	2,998.13	321.85	759.23	818.88	2.646	-2.022	1.859	
3,514.77	61.86	89.99	3,015.17	322.15	793.49	849.05	2.749	-2.665	0.753	
3,555.95	57.08	91.40	3,036.08	321.73	828.94	879.93	3.593	-3.482	1.027	
3,596.23	51.69	92.76	3,059.53	320.55	861.65	908.04	4.098	-4.014	1.013	
3,638.07	48.65	94.05	3,086.33	318.65	893.72	935.23	2.292	-2.180	0.925	
3,678.39	49.69	93.65	3,112.69	316.60	924.16	960.93	0.806	0.774	-0.298	
3,718.84	50.17	93.80	3,138.73	314.59	955.05	987.04	0.366	0.356	0.111	

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Wellbore:	F-14	Survey Calculation Method:	Minimum Curvature
Design:	F-14	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
3,729.86	50.26	93.74	3,145.78	314.04	963.50	994.17	0.275	0.245	-0.163	
3,750.00	50.26	93.74	3,158.65	313.03	978.96	1,007.23	0.000	0.000	0.000	

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Design Targets									
Target Name									
- hit/miss target	Dip Angle	Dip Dir.	TVD	+N/-S	+E/-W	Northing	Easting	Latitude	Longitude
- Shape	(°)	(°)	(m)	(m)	(m)	(m)	(m)		
F14 stying Shallow	0.00	0.00	2,823.57	248.56	347.10	6,478,812.00	435,397.00	58° 26' 38.0241 N	1° 53' 36.0678 E
- actual wellpath misses target center by 46.93m at 3013.50m MD (2865.90 TVD, 245.42 N, 327.07 E)									
- Point									
F14 stying: Top 060307	0.00	0.00	2,834.90	256.57	365.11	6,478,820.00	435,415.00	58° 26' 38.2923 N	1° 53' 37.1695 E
- actual wellpath misses target center by 44.51m at 3038.73m MD (2875.97 TVD, 255.30 N, 347.98 E)									
- Point									
F14 stying Deep	0.00	0.00	2,840.90	261.57	376.11	6,478,825.00	435,426.00	58° 26' 38.4598 N	1° 53' 37.8426 E
- actual wellpath misses target center by 43.49m at 3052.18m MD (2881.06 TVD, 260.07 N, 359.47 E)									
- Point									
F14 T1 Rev6	4.00	300.00	2,842.90	231.86	289.08	6,478,795.30	435,339.00	58° 26' 37.4534 N	1° 53' 32.5088 E
- actual wellpath misses target center by 7.81m at 2967.73m MD (2845.84 TVD, 224.99 N, 291.37 E)									
- Rectangle (sides W55.40 H49.22 D0.00)									
F14 RTD Target	0.00	0.00	2,842.90	233.87	324.35	6,478,797.31	435,374.26	58° 26' 37.5372 N	1° 53' 34.6807 E
- actual wellpath misses target center by 19.67m at 2996.95m MD (2858.94 TVD, 238.27 N, 313.85 E)									
- Rectangle (sides W38.00 H64.00 D0.00)									
F-14 rev6_target1_Top I	4.00	300.00	2,842.90	231.86	289.08	6,478,795.30	435,339.00	58° 26' 37.4534 N	1° 53' 32.5088 E
- actual wellpath misses target center by 7.81m at 2967.73m MD (2845.84 TVD, 224.99 N, 291.37 E)									
- Polygon									
Point 1			2,840.41	-9.04	35.93	6,478,786.31	435,374.84		
Point 2			2,841.91	-36.75	-4.75	6,478,758.58	435,334.22		
Point 3			2,845.39	9.04	-35.93	6,478,804.29	435,303.16		
Point 4			2,843.89	36.75	4.75	6,478,832.02	435,343.78		
F14 RTD Target 250907	0.00	0.00	2,842.90	233.56	323.09	6,478,797.00	435,373.00	58° 26' 37.5265 N	1° 53' 34.6034 E
- actual wellpath misses target center by 19.03m at 2995.80m MD (2858.45 TVD, 237.76 N, 312.95 E)									
- Point									
F14 Revised 260307	0.00	0.00	2,845.90	236.56	330.09	6,478,800.00	435,380.00	58° 26' 37.6272 N	1° 53' 35.0319 E
- actual wellpath misses target center by 19.85m at 3003.96m MD (2861.92 TVD, 241.36 N, 319.40 E)									
- Point									
Volve P50 Topp	0.00	0.94	2,859.00	96.51	266.07	6,478,660.00	435,316.00	58° 26' 33.0675 N	1° 53' 31.2283 E
- actual wellpath misses target center by 113.05m at 2906.45m MD (2813.41 TVD, 198.17 N, 246.91 E)									
- Point									
F14 stying: flt	0.00	0.00	2,924.90	336.59	545.17	6,478,900.00	435,595.00	58° 26' 40.9740 N	1° 53' 48.1864 E
- actual wellpath misses target center by 31.28m at 3253.43m MD (2933.46 TVD, 306.58 N, 547.15 E)									
- Point									
F14 Base Hugin Upper E	4.00	300.00	2,944.90	305.58	615.19	6,478,869.00	435,665.00	58° 26' 40.0090 N	1° 53' 52.5337 E
- actual wellpath misses target center by 6.93m at 3322.06m MD (2946.18 TVD, 312.33 N, 614.31 E)									
- Rectangle (sides W24.61 H27.70 D0.00)									
F14 Guide Point Fault	0.00	0.00	2,985.22	322.59	755.24	6,478,886.00	435,805.00	58° 26' 40.6327 N	1° 54' 1.1484 E
- actual wellpath misses target center by 10.15m at 3467.75m MD (2994.51 TVD, 321.68 N, 751.26 E)									
- Point									
Guidepoint Fault Fallbac	0.00	0.00	2,994.90	322.59	755.24	6,478,886.00	435,805.00	58° 26' 40.6327 N	1° 54' 1.1484 E
- actual wellpath misses target center by 1.52m at 3471.59m MD (2996.09 TVD, 321.76 N, 754.76 E)									
- Point									
F14 Stying : TD	0.00	0.00	2,994.90	374.61	683.22	6,478,938.00	435,733.00	58° 26' 42.2755 N	1° 53' 56.6566 E
- actual wellpath misses target center by 61.77m at 3407.67m MD (2971.99 TVD, 318.61 N, 695.67 E)									
- Point									
Volve P51 Bunn	0.00	0.94	3,016.02	578.68	423.63	6,479,142.00	435,473.50	58° 26' 48.7323 N	1° 53' 40.4499 E
- actual wellpath misses target center by 303.34m at 3194.60m MD (2923.05 TVD, 297.66 N, 489.97 E)									
- Point									
F14 Base Hugin	0.00	0.00	3,081.90	317.09	871.08	6,478,880.50	435,920.80	58° 26' 40.5160 N	1° 54' 8.2937 E

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Site:	Volve F	MD Reference:	Inspirer Rotary Table @ 54.90m (Actual Rotary Table)
Well:	F-14	North Reference:	Grid
Wellbore:	F-14	Survey Calculation Method:	Minimum Curvature
Design:	F-14	Database:	Production EDM P246N

- actual wellpath misses target center by 11.69m at 3617.95m MD (3073.23 TVD, 319.65 N, 878.48 E)									
- Point									
Block 2 Base Hugin 1404	0.00	0.00	3,107.90	315.59	918.30	6,478,879.00	435,968.00	58° 26' 40.4925 N	1° 54' 11.2054 E
- actual wellpath misses target center by 1.40m at 3670.88m MD (3107.82 TVD, 316.97 N, 918.46 E)									
- Point									
Base Hugin Rev 5 2204	26.00	62.00	3,129.90	312.97	943.64	6,478,876.38	435,993.33	58° 26' 40.4211 N	1° 54' 12.7697 E
- actual wellpath misses target center by 2.42m at 3704.52m MD (3129.54 TVD, 315.32 N, 944.09 E)									
- Polygon									
Point 1			3,133.19	-23.95	21.23	6,478,852.08	436,013.88		
Point 2			3,118.71	-19.86	-18.34	6,478,857.74	435,977.28		
Point 3			3,129.30	23.19	-13.89	6,478,899.63	435,979.57		
Point 4			3,143.77	19.10	25.68	6,478,893.97	436,016.17		
Rev-6 Base Hugin	26.00	62.00	3,129.90	312.59	943.31	6,478,876.00	435,993.00	58° 26' 40.4087 N	1° 54' 12.7498 E
- actual wellpath misses target center by 2.85m at 3704.29m MD (3129.39 TVD, 315.33 N, 943.92 E)									
- Rectangle (sides W39.78 H43.28 D0.00)									

Casing Points					
	Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)
	196.80	196.80	30"	30.000	36.000
	1,076.90	1,076.46	20"	20.000	26.000
	1,604.00	1,603.55	10 3/4"	10.750	12.250
	2,275.40	2,274.05	14"	14.000	17.500
	2,783.50	2,728.26	9 5/8"	9.625	12.250
	3,695.00	3,123.41	7"	7.000	8.500

Checked By: _____	Approved By: _____	Date: _____
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